local-1777880023450

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Engineering report

Generated

Project ID	**local-1777880023450**
Domain	**hvac_and_refrigeration**
Engine	**PDF Engine v2 (One-Shot)**

TOTALS AT A GLANCE

MODULES	BOM ROWS	UNIT COST
5	**36**	**£1,021.80**

CEILING	HEADROOM
None	**N/A**

Contents

1. Brief

Subject

Space heating and low-temp hot water generation for small commercial buildings.

Constraints

Unit Cost Ceiling	**£0.00**
Max Mass	**-**
Target Process	**Vapour compression refrigeration cycle (air-to-water) driven by electrical inverter power.**
Target Material	**R290 Refrigerant, 316 Stainless Steel (Condenser), Aluminum/Copper (Evaporator), Galvanized Steel (Chassis).**
Tolerance	**Capacity: >= 30kW thermal (+/- 5%). COP: >= 3.5 at A2/W35. Noise: <65dB at 1 meter.**
Quantity	**500 units per year with a Target Cost (BOM/Mfg) of £4,500.**

Compliance Notes

Must comply with MCS 007, EN 378 for A3 refrigerants, UKCA marking, and adhere strictly to 1200x800x1400mm dimensions.

2. Regulatory posture

BS EN 378

not-started

Refrigerating systems and heat pumps. Safety and environmental requirements

Covers safety classifications, charge limits, and design considerations for toxic and flammable refrigerants.

APPLICABILITY

Mandatory since R290 is classified as an A3 highly flammable refrigerant.

DESIGN IMPACT

Requires system to be designed as a monobloc to prevent pumping A3 refrigerant indoors above acceptable charge limits.

EVIDENCE REQUIRED

F-Gas / refrigerant compliance dossier and safety calculations

OWNER

Mechanical Engineer

GAP ACTION

Re-architect split system to a water-coupled monobloc system.

IEC 60335-2-40

not-started

Safety of household and similar electrical appliances - Particular requirements for electrical heat pumps, air-conditioners and dehumidifiers

Establishes electrical and thermal safety procedures for motor-compressor equipment.

APPLICABILITY

Mandatory for CE/UKCA marking of electrically driven heat pumps.

DESIGN IMPACT

Requires sealed electrical boxes (IP54 or greater), ignition-free switching, and safe clearance from combustible components.

EVIDENCE REQUIRED

CB test report from accredited laboratory (e.g. TUV, Intertek).

OWNER

Compliance Engineer

GAP ACTION

Initiate electrical pre-compliance review focusing on ATEX/ignition-free zones.

EN 14511

not-started

Air conditioners, liquid chilling packages and heat pumps for space heating and cooling

Standardized test conditions for establishing thermal capacity and COP.

APPLICABILITY

Mandatory for proving 30kW and 3.5 COP claims.

DESIGN IMPACT

Determines the exact lab testing temperatures (A2/W35) the thermodynamic cycle must be optimized against.

EVIDENCE REQUIRED

Performance test report from an ISO 17025 accredited psychrometric chamber.

OWNER

Thermodynamics Engineer

GAP ACTION

Build prototype and schedule chamber testing.

EN 14825

not-started

Testing and rating at part load conditions and calculation of seasonal performance

Calculates the SCOP (Seasonal Coefficient of Performance) for EcoDesign regulatory requirements.

APPLICABILITY

Mandatory for legal sale in the UK (ErP directive).

DESIGN IMPACT

Requires the unit to have part-load modulation capability. Inverter drive logic must be highly tuned.

EVIDENCE REQUIRED

SCOP calculation report from lab test data.

OWNER

Control Systems Engineer

GAP ACTION

Develop inverter and EEV mapping for 30%, 50%, and 75% load conditions.

MCS 007

not-started

Heat Pump Product Certification Scheme Requirements

UK specific scheme ensuring quality, performance, and durability for accessing government grants.

APPLICABILITY

Crucial requirement for the specified UK commercial market.

DESIGN IMPACT

Restricts design modifications post-certification and demands strict adherence to factory production control (ISO 9001).

EVIDENCE REQUIRED

MCS Product Certificate and Factory Audit Report.

OWNER

Quality Assurance Manager

GAP ACTION

Engage an MCS accredited certification body (e.g., BRE or KIWA).

3. Sizing optimisation

INFEASIBLE — design constraints not satisfied.

Envelope

Kind	**generic_room**
Dimensions	**5000 × 5000 × 3000 mm**
Floor area	**25 m²**

Module Dimensions

MODULE	W × D × H (MM)	AREA M²	MOUNT	SCALED BY
sealed_r290_circuit	2236 × 2236 × 2400	5	floor	mass_proportional
airflow_assembly	2236 × 2236 × 2400	5	floor	mass_proportional
electrical_and_controls	2236 × 2236 × 2400	5	floor	mass_proportional
chassis_structural_frame	2236 × 2236 × 2400	5	floor	mass_proportional
indoor_hydrobox	2236 × 2236 × 2400	5	floor	mass_proportional

Conflicts

- Total module footprint (25.0m2) exceeds envelope budget (21.3m2)

Recommendations

- Consider a larger envelope
- Optimize module footprint

4.1. Sealed R290 Refrigeration Circuit

Lead: 12 weeks

Purpose

Executes the vapour compression cycle to extract ambient heat and transfer it to the hydronic loop.

Why it matters

Meeting the aggressive >= 3.5 COP threshold at the challenging A2/W35 conditions relies entirely on the efficiency, component sizing, and EEV tuning of this exact module.

Description

The highly-tuned thermodynamic engine of the monobloc heat pump. Because R290 is an A3 highly flammable refrigerant, this entire circuit is factory-sealed and isolated outside to satisfy BS EN 378 charge limits. The circuit forces low-temp R290 gas to high pressure via the specific low-GWP scroll compressor, subsequently rejecting concentrated heat into the commercial hydronic loop through the stainless steel BPHE condenser.

KEY PARTS

- Inverter-driven scroll compressor
- Oversized aluminum/copper evaporator
- 316 SS Brazed Plate Heat Exchanger (BPHE)
- Electronic Expansion Valve (EEV)

FAILURE MODES

- Refrigerant leak at braze points
- Evaporator coil frosting
- Compressor mechanical failure

OPEN QUESTIONS

- Exact calculated R290 charge mass (estimated 2.5-4kg)
- Optimal precise defrost cycle timing

Cost Breakdown

PART #	NAME	TYPE	COST
sealed_r290_circuit-001	Inverter-driven scroll compressor	COTS	£20.00
sealed_r290_circuit-002	Oversized aluminum/copper evaporator	COTS	£20.00
sealed_r290_circuit-003	316 SS Brazed Plate Heat Exchanger (BPHE)	COTS	£20.00
sealed_r290_circuit-004	Electronic Expansion Valve (EEV)	COTS	£20.00
sealed_r290_circuit-005	Refrigerant grade copper piping kit	COTS	£45.00
sealed_r290_circuit-006	Closed-cell elastomeric pipe insulation	COTS	£12.50
sealed_r290_circuit-007	High-pressure safety transducer	COTS	£28.00
		Module Total:	**£165.50**

4.2. Airflow and Acoustic Assembly

Lead: 8 weeks

Purpose

Drives massive volumes of ambient air across the evaporator coil while maintaining strict acoustic limits.

Why it matters

Insufficient airflow instantly degrades capacity and degrades COP, while excessive RPM breaks the mandatory acoustic limits required for legal sale and MCS 007 certification.

Description

A high-volume air movement system designed to overcome the static pressure of an oversized, frost-prone evaporator coil. Given the massive >10,000 m^3/hr air volume required for a 30kW load, this module mandates low-RPM, Electronically Commutated (EC) axial fans leveraging swept blade tech alongside structural dampening to stay under the 65dB MCS noise limit.

KEY PARTS

- Large diameter (710mm) EC axial fan
- Swept composite fan blades
- Acoustic composite fan shroud
- Compressor acoustic jacket

FAILURE MODES

- Fan motor bearing wear
- Ice buildup unbalancing fan blades
- Excessive low-frequency hum/resonance

OPEN QUESTIONS

- Actual air pressure drop across frosted evaporator state
- Low-frequency noise propagation profile

Cost Breakdown

PART #	NAME	TYPE	COST
airflow_assembly-001	Large diameter (710mm) EC axial fan	COTS	£20.00
airflow_assembly-002	Swept composite fan blades	COTS	£20.00
airflow_assembly-003	Acoustic composite fan shroud	COTS	£20.00
airflow_assembly-004	Compressor acoustic jacket	COTS	£20.00
airflow_assembly-005	Steel fan grille/finger guard	COTS	£18.00
airflow_assembly-006	M8 stainless steel nyloc fastener set	COTS	£5.50
airflow_assembly-007	Anti-vibration fan mounting grommets	COTS	£4.20
		Module Total:	**£107.70**

4.3. Electrical and Controls Enclosure

Lead: 10 weeks

Purpose

Manages power distribution, drives modulation algorithms, and ensures safe control matching IEC 60335-2-40.

Why it matters

It implements the part-load logic to meet EcoDesign (ErP) requirements and ensures absolute electrical safety against spark environments, required for CE/UKCA certification.

Description

The decision-making and power delivery hub. Because the equipment handles an A3 flammable refrigerant, the enclosure strictly segregates all electronics from the refrigeration bay. It contains ignition-free, ATEX-compatible relays and an inverter drive programmed to perfectly modulate the 30kW cycle, crucial for EN 14825 SCOP performance.

KEY PARTS

- IP54+ rated sealed enclosure
- System control mainboard
- Inverter drive module
- Solid-state ATEX relays

FAILURE MODES

- Inverter overheating
- PCB condensation and shorting
- Faulty electronic valve control

OPEN QUESTIONS

- Inverter drive EMI/RFI profile
- Required cooling capacity for the sealed electrical box

Cost Breakdown

PART #	NAME	TYPE	COST
electrical_and_controls-001	IP54+ rated sealed enclosure	Custom	£50.00
electrical_and_controls-002	System control mainboard	COTS	£20.00
electrical_and_controls-003	Inverter drive module	COTS	£20.00
electrical_and_controls-004	Solid-state ATEX relays	COTS	£20.00
electrical_and_controls-005	IP68 nylon cable gland kit	COTS	£8.00
electrical_and_controls-006	ATEX compliant power wiring harness	COTS	£35.00
electrical_and_controls-007	Flammable gas A3 warning labels	COTS	£2.00
electrical_and_controls-008	PCB mounting standoff kit	COTS	£3.00
		Module Total:	**£158.00**

4.4. Chassis and Structural Frame

Lead: 6 weeks

Purpose

Provides physical support, environmental protection, and vibration handling within the 1200x800x1400mm footprint.

Why it matters

It sets the physical limits of the system, dictates the bulk assembly and logistics costs, and prevents rapid corrosion in long-term commercial installations.

Description

The rigid external shell and internal skeleton of the monobloc unit, strictly confined to the specified 1200x800x1400mm envelope. To meet the ultra-lean £4,500 target BOM constraint at 500 units/year, it foregoes injection molding in favor of robust, CNC-folded sheet metal using off-the-shelf profiles—potentially contract-manufactured in Eastern Europe or Southeast Asia.

KEY PARTS

- CNC-folded galvanized sheet metal
- Powder-coated outer panels
- Elastomeric anti-vibration mounts
- Structural aluminum support struts

FAILURE MODES

- Corrosion fatigue
- Panel rattling transferring acoustic noise
- Frame warping during shipping

OPEN QUESTIONS

- Exact overall assembled mass
- Resonant frequencies of the full metal assembly

Cost Breakdown

PART #	NAME	TYPE	COST
chassis_structural_frame-001	CNC-folded galvanized sheet metal	COTS	£20.00
chassis_structural_frame-002	Powder-coated outer panels	COTS	£20.00
chassis_structural_frame-003	Elastomeric anti-vibration mounts	COTS	£20.00
chassis_structural_frame-004	Structural aluminum support struts	COTS	£20.00
chassis_structural_frame-005	EPDM perimeter weather seal gasket	COTS	£14.00
chassis_structural_frame-006	M6 blind rivet nut (Rivnut) pack	COTS	£9.00
chassis_structural_frame-007	M10 forged lifting eye bolts	COTS	£12.00
		Module Total:	**£115.00**

4.5. Indoor Hydrobox Assembly

Lead: 8 weeks

Purpose

Transfers thermal capacity from the monobloc to the indoor commercial distribution system without bringing refrigerant indoors.

Why it matters

This exact sub-assembly satisfies the fundamental pivot required by the BS EN 378 safety mandate, eliminating A3 refrigerant poisoning or explosive hazards inside the commercial building.

Description

The critical compliance bridge for this specific system. By confining this module entirely to hydronic components and keeping it inside the building, the highly flammable R290 remains outdoors. The hydrobox houses circulation pumps capable of commercial loop pressure, a backup immersion heater for extreme cold snaps, and expansion tanks.

KEY PARTS

- Circulating pump
- 10L expansion vessel
- Electric backup immersion heater
- Local User Interface (HMI)

FAILURE MODES

- Pump cavitation
- Immersion heater scale buildup and burnout
- Internal water leaks

OPEN QUESTIONS

- Required pump 'head' sizing for varying length commercial pipe runs
- Acceptable voltage draw for the backup immersion element

Cost Breakdown

PART #	NAME	TYPE	COST
indoor_hydrobox-001	Circulating pump	COTS	£20.00
indoor_hydrobox-002	10L expansion vessel	COTS	£20.00
indoor_hydrobox-003	Electric backup immersion heater	COTS	£20.00
indoor_hydrobox-004	Local User Interface (HMI)	COTS	£20.00
indoor_hydrobox-005	3-Bar Pressure Relief Valve (PRV)	COTS	£16.50
indoor_hydrobox-006	Inline vortex flow sensor	COTS	£32.00
indoor_hydrobox-007	Expansion vessel mounting bracket	Custom	£6.50
		Module Total:	**£135.00**

5. BOM master — 36 rows

PART #	NAME	MODULE	TYPE	MASS	COST
sealed_r290_circuit-001	Inverter-driven scroll compressor varies	Sealed R290 Refrigeration Circuit	COTS	2.5kg	£20.00
sealed_r290_circuit-002	Oversized aluminum/copper evaporator varies	Sealed R290 Refrigeration Circuit	COTS	2.5kg	£20.00
sealed_r290_circuit-003	316 SS Brazed Plate Heat Exchanger (BPHE) varies	Sealed R290 Refrigeration Circuit	COTS	2.5kg	£20.00
sealed_r290_circuit-004	Electronic Expansion Valve (EEV) varies	Sealed R290 Refrigeration Circuit	COTS	2.5kg	£20.00
airflow_assembly-001	Large diameter (710mm) EC axial fan varies	Airflow and Acoustic Assembly	COTS	2.5kg	£20.00
airflow_assembly-002	Swept composite fan blades varies	Airflow and Acoustic Assembly	COTS	2.5kg	£20.00
airflow_assembly-003	Acoustic composite fan shroud varies	Airflow and Acoustic Assembly	COTS	2.5kg	£20.00
airflow_assembly-004	Compressor acoustic jacket varies	Airflow and Acoustic Assembly	COTS	2.5kg	£20.00
electrical_and_controls-001	IP54+ rated sealed enclosure varies	Electrical and Controls Enclosure	Custom	2.5kg	£50.00
electrical_and_controls-002	System control mainboard varies	Electrical and Controls Enclosure	COTS	2.5kg	£20.00
electrical_and_controls-003	Inverter drive module varies	Electrical and Controls Enclosure	COTS	2.5kg	£20.00
electrical_and_controls-004	Solid-state ATEX relays varies	Electrical and Controls Enclosure	COTS	2.5kg	£20.00
chassis_structural_frame-001	CNC-folded galvanized sheet metal varies	Chassis and Structural Frame	COTS	2.5kg	£20.00
chassis_structural_frame-002	Powder-coated outer panels varies	Chassis and Structural Frame	COTS	2.5kg	£20.00
chassis_structural_frame-003	Elastomeric anti-vibration mounts varies	Chassis and Structural Frame	COTS	2.5kg	£20.00
chassis_structural_frame-004	Structural aluminum support struts varies	Chassis and Structural Frame	COTS	2.5kg	£20.00
indoor_hydrobox-001	Circulating pump varies	Indoor Hydrobox Assembly	COTS	2.5kg	£20.00
indoor_hydrobox-002	10L expansion vessel varies	Indoor Hydrobox Assembly	COTS	2.5kg	£20.00

PART #	NAME	MODULE	TYPE	MASS	COST
indoor_hydrobox-003	Electric backup immersion heater varies	Indoor Hydrobox Assembly	COTS	2.5kg	£20.00
indoor_hydrobox-004	Local User Interface (HMI) varies	Indoor Hydrobox Assembly	COTS	2.5kg	£20.00
sealed_r290_circuit-005	Refrigerant grade copper piping kit Copper	Sealed R290 Refrigeration Circuit	COTS	3.5kg	£45.00
sealed_r290_circuit-006	Closed-cell elastomeric pipe insulation EPDM Foam	Sealed R290 Refrigeration Circuit	COTS	0.8kg	£12.50
sealed_r290_circuit-007	High-pressure safety transducer Brass/Electronics	Sealed R290 Refrigeration Circuit	COTS	0.1kg	£28.00
airflow_assembly-005	Steel fan grille/finger guard Powder-coated steel	Airflow and Acoustic Assembly	COTS	1.2kg	£18.00
airflow_assembly-006	M8 stainless steel nyloc fastener set 304 Stainless Steel	Airflow and Acoustic Assembly	COTS	0.4kg	£5.50
airflow_assembly-007	Anti-vibration fan mounting grommets Silicone Rubber	Airflow and Acoustic Assembly	COTS	0.1kg	£4.20
electrical_and_controls-005	IP68 nylon cable gland kit Nylon	Electrical and Controls Enclosure	COTS	0.2kg	£8.00
electrical_and_controls-006	ATEX compliant power wiring harness Copper/TPE	Electrical and Controls Enclosure	COTS	1.5kg	£35.00
electrical_and_controls-007	Flammable gas A3 warning labels Vinyl	Electrical and Controls Enclosure	COTS	0.05kg	£2.00
electrical_and_controls-008	PCB mounting standoff kit Nylon	Electrical and Controls Enclosure	COTS	0.05kg	£3.00
chassis_structural_frame-005	EPDM perimeter weather seal gasket EPDM Rubber	Chassis and Structural Frame	COTS	1.1kg	£14.00
chassis_structural_frame-006	M6 blind rivet nut (Rivnut) pack Zinc Plated Steel	Chassis and Structural Frame	COTS	0.6kg	£9.00
chassis_structural_frame-007	M10 forged lifting eye bolts Forged Steel	Chassis and Structural Frame	COTS	0.8kg	£12.00
indoor_hydrobox-005	3-Bar Pressure Relief Valve (PRV) Brass	Indoor Hydrobox Assembly	COTS	0.4kg	£16.50
indoor_hydrobox-006	Inline vortex flow sensor Polymer	Indoor Hydrobox Assembly	COTS	0.3kg	£32.00
indoor_hydrobox-007	Expansion vessel mounting bracket Galvanized Steel	Indoor Hydrobox Assembly	Custom	0.9kg	£6.50

Total Unit Cost: **£681.20**

6. Cost waterfall

UNIT COST	CEILING	HEADROOM
£1,021.80	None	N/A

Per-Module Roll-up

MODULE	% OF UNIT	COST
Module sealed_r290_circuit	24.3%	£248.25
Module airflow_assembly	15.8%	£161.55
Module electrical_and_controls	23.2%	£237.00
Module chassis_structural_frame	16.9%	£172.50
Module indoor_hydrobox	19.8%	£202.50

NRE Total	**£3,000.00**
Overhead Multiplier	**1.5×**

Grand Total (Unit + NRE) **£4,021.80**

7. Risks register

Sealed R290 Refrigeration Circuit

A3 refrigerant leak causing potential ignition in enclosed zones

S:5 × L:2 = 10

CONTROLS	MITIGATION
None	Minimize mechanical braze joints, use high-strength piping, isolate from electrical ignition sources.

Owner: Mechanical Engineer

Severe evaporator frosting at A2/W35 dropping capacity below 30kW limit

S:4 × L:4 = 16

CONTROLS	MITIGATION
None	Implement aggressive hot-gas bypass or optimized reverse-cycle defrost.

Owner: Thermodynamics Engineer

Airflow and Acoustic Assembly

System exceeds 65dB noise limit at 1m

S:4 × L:3 = 12

CONTROLS	MITIGATION
None	Integrate specific Ziehl-Abegg/ebm-papst ultra-low noise fans and validate acoustic jacket sizing.

Owner: Acoustic Engineer

Electrical and Controls Enclosure

Relay arc ignites localized R290 gas buildup

S:5 × L:1 = 5

CONTROLS	MITIGATION
None	Ensure electrical enclosure meets IP54 seal and utilize exclusively solid-state/spark-free switching.

Owner: Compliance Engineer

Chassis and Structural Frame

Sheet metal fabrication and finishing pushes unit BOM beyond £4,500

S:4 × L:3 = 12

CONTROLS	MITIGATION
None	Utilize pure CNC off-the-shelf profiles without custom tooling, outsource base structural fabrication.

Owner: Manufacturing Engineer

Indoor Hydrobox Assembly

Standard pump head insufficient for sprawling commercial hot water distributions

S:3 × L:4 = 12

CONTROLS	MITIGATION
None	Define max internal pipe runs in documentation or offer high-torque pump upgrades.

Owner: Systems Engineer

8. Supplier shortlist — 36 parts

Inverter-driven scroll compressor (sealed_r290_circuit-001)

Oversized aluminum/copper evaporator (sealed_r290_circuit-002)

316 SS Brazed Plate Heat Exchanger (BPHE) (sealed_r290_circuit-003)

Electronic Expansion Valve (EEV) (sealed_r290_circuit-004)

Large diameter (710mm) EC axial fan (airflow_assembly-001)

Swept composite fan blades (airflow_assembly-002)

Acoustic composite fan shroud (airflow_assembly-003)

Compressor acoustic jacket (airflow_assembly-004)

IP54+ rated sealed enclosure (electrical_and_controls-001)

System control mainboard (electrical_and_controls-002)

Inverter drive module (electrical_and_controls-003)

Solid-state ATEX relays (electrical_and_controls-004)

CNC-folded galvanized sheet metal (chassis_structural_frame-001)

Powder-coated outer panels (chassis_structural_frame-002)

Elastomeric anti-vibration mounts (chassis_structural_frame-003)

Structural aluminum support struts (chassis_structural_frame-004)

Circulating pump (indoor_hydrobox-001)

10L expansion vessel (indoor_hydrobox-002)

Electric backup immersion heater (indoor_hydrobox-003)

Local User Interface (HMI) (indoor_hydrobox-004)

Refrigerant grade copper piping kit (gap-fill-1777880139370-0)

Closed-cell elastomeric pipe insulation (gap-fill-1777880139370-1)

High-pressure safety transducer (gap-fill-1777880139370-2)

Steel fan grille/finger guard (gap-fill-1777880139370-3)

M8 stainless steel nyloc fastener set (gap-fill-1777880139370-4)

Anti-vibration fan mounting grommets (gap-fill-1777880139370-5)

IP68 nylon cable gland kit (gap-fill-1777880139370-6)

ATEX compliant power wiring harness (gap-fill-1777880139370-7)

Flammable gas A3 warning labels (gap-fill-1777880139370-8)

PCB mounting standoff kit (gap-fill-1777880139370-9)

EPDM perimeter weather seal gasket (gap-fill-1777880139370-10)

M6 blind rivet nut (Rivnut) pack (gap-fill-1777880139370-11)

M10 forged lifting eye bolts (gap-fill-1777880139370-12)

3-Bar Pressure Relief Valve (PRV) (gap-fill-1777880139370-13)

Inline vortex flow sensor (gap-fill-1777880139370-14)

Expansion vessel mounting bracket (gap-fill-1777880139370-15)

9. Audit log

> Pipeline execution log — generated by PDF Engine v2.

This document was generated automatically. Contents represent the synthesised output of the specialist pipeline.